

The Squeeze-Thermal State: A Type-I No-Go, the Scale-Line Construction, and the Type III₁ Verdict

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Abstract. A companion paper formulates the state problem this series' preparation phase converges on: on the CCR structure produced by the two-contraction tower, find a state invariant under the surviving $SO(3)$ remnant and KMS with respect to the squeeze flow — the $\hbar$ -preserving shadow of the octonionic grading dilation. We solve this problem in outline, with every load-bearing step proved, computationally verified, or cited and flagged. First, a no-go theorem: no such state — indeed no squeeze-invariant normal state at all — exists on any finite number of CCR fibers, and no invariant quasi-free covariance exists for the diagonal squeeze on any number of static fibers; the obstruction runs exactly through $W \neq 0$ and evaporates classically. Type-III necessity is thereby proven inside the model rather than imported. Second, the escape: the fiber squeeze is unitarily conjugate to translations along the log-radial line — the scale continuum itself becomes the mode space of second quantization, with spectrum absolutely continuous and equal to $\mathbb{R}$, forced rather than chosen. Third, the construction: a chiral, current-type quasi-free β -KMS state for translations over the scale line, rotation-invariant automatically, with $W = \hbar \cdot 1$ uniform. Fourth, the type: the closed group generated by the modular spectral data is computed directly — the forced interval spectrum saturates $\mathbb{R}$, while the counterfactual state coupled to the discrete grading ladder, run on the octonionic weights $(1, 2, 3)$, freezes on a lattice, giving type III $_{\lambda}$ with $\lambda = e^{-\beta E_0}$ — the contraction scale imprinted. Conditional on one standard, cited link (the centralizer condition for free thermal states), the factor of the squeeze-thermal state is the unique hyperfinite type III₁.

1. Introduction

The companion paper (*The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the $SO(3)$ Remnant of G_2*) ends by posing a precise problem. The two-contraction tower terminates in the $3+3+1$ Heisenberg algebra with $\{X^a, P_b\} = -2\delta_b^a W$ and W central; by the centrality theorem of this series and Schur's lemma, $W = \hbar \cdot 1$ in any irreducible representation, fixing the CCR fiber. Two structures are inherited from the octonionic level: the squeeze flow — the unique $\hbar$ -preserving component of the grading dilation, with generator $D = \frac{1}{2} \sum_a (X_a P_a + P_a X_a)$ — and the $\mathfrak{so}(3)$ remnant of G_2 , acting as one rotation on X and P simultaneously. The state problem: find ω that is KMS for the squeeze flow and invariant under the remnant. Three structural theorems shape the problem from outside (all cited in the companion paper): dynamics must be state-borne (Alfsen–Shultz), no algebra-resident density can generate the flow at a type III target (Takesaki), and the type of the resulting factor is a computation, not a premise.

This note solves the problem in outline. Statuses are labeled throughout: *proved* (here),

verified (computationally, this series' standard), *standard* (textbook technology), *cited* (external result, flagged for verification), *open*.

2. The conjugacy: the squeeze is translation along the scale line

Proposition 1 (proved; verified). Let $V : L^2(\mathbb{R}_+, dr) \rightarrow L^2(\mathbb{R}, du)$ be $(Vf)(u) = e^{u/2} f(e^u)$, $u = \log r$. Then V is unitary and intertwines the squeeze flow $(U_t f)(r) = e^{t/2} f(e^t r)$ with rigid translations $u \mapsto u + t$. On the full fiber the same mechanism holds with the three-dimensional Jacobian: $W : L^2(\mathbb{R}^3) \rightarrow L^2(\mathbb{R}_u) \otimes L^2(S^2)$, $(Wf)(u, \theta) = e^{3u/2} f(e^u \theta)$, is unitary for the measure $r^2 dr d\Omega$ and carries the squeeze $(e^{itD} f)(x) = e^{3t/2} f(e^t x)$ to translation $u \mapsto u + t$ identically on every angular channel, with generator $-i\partial_u$. The translated line is the *radial scale* coordinate $u = \log r$, not a Cartesian axis of the Heisenberg fiber: what the flow shifts is the continuum of scales.

Verification. The intertwining is an exact change-of-variables identity, confirmed numerically to defect 1.2×10^{-15} ; unitarity (the Jacobian half-power) confirmed by quadrature, $\|Vf\|^2 = 0.7106535$ against $\|f\|^2 = 0.7106546$, the difference being finite-window tail truncation.

Corollary 1.1. The spectrum of D is purely absolutely continuous and equal to $\mathbb{R}$ — forced by the conjugacy, not chosen.

Corollary 1.2 (verified). The squeeze commutes with the diagonal $SO(3)$ exactly: at the symplectic level both are block-scalar, and $\|[S_t, R \oplus R]\| = 0$ to machine precision. Rotations act purely on the angular factor.

3. The no-go: the state cannot exist in type I, and W is the reason

Proposition 2 (proved). On any finite number of fibers, no normal state is invariant under the squeeze flow; a fortiori, no KMS state exists there.

Proof. A normal invariant state is a positive trace-one operator ρ commuting with e^{itD} . Trace-one implies compact, so ρ 's spectrum away from zero is pure point with finite-rank spectral projections; these commute with e^{itD} , yielding finite-dimensional invariant subspaces on which D has eigenvalues — contradicting Corollary 1.1 (the total generator over n fibers retains purely absolutely continuous spectrum). Hence $\rho = 0$, contradicting trace one. KMS implies invariance, so no KMS state. ■

Proposition 3 (proved; verified). For the diagonal squeeze acting on any number n of static fibers, no invariant quasi-free covariance exists. The space of invariant symmetric forms was solved exactly (dimension 1 for $n = 1$, 9 for $n = 3$, all cross-pairings included): every solution has identically vanishing x - x block. The quasi-free positivity bound $\eta(x, x) \eta(p, p) \geq \frac{1}{4} \sigma(x, p)^2$, with $\sigma(x, p) = -2W$, then reads $0 \geq W^2$: violated for every $W \neq 0$, and vacuous exactly at $W = 0$.

Remark (the mechanism). The no-go is strictly quantum. The structural $\hbar$ — whose

survival through two contractions this series engineered — is itself what expels its own thermal flow from every type-I setting. This is the internal counterpart of Takesaki's criterion (semifinite $\iff$ modular flow inner): the sought state can exist only where density matrices fail. Proposition 3 also identifies the escape route: invariance fails for any flow that *squeezes static modes*; it can hold only for a flow that *translates modes along a continuum* — and Proposition 1 shows the fiber supplies exactly that continuum, the scale line itself.

4. The construction

Take the one-particle space $H_1 = L^2(\mathbb{R}_u) \otimes L^2(S^2)$ with one-particle generator $h = -i\partial_u$ (Proposition 1). Perform the chiral positive-frequency restriction — the $p > 0$ half of each translation line, the $p < 0$ half entering as conjugates [standard CCR technology for generators with two-sided spectrum]. Handle the $p \rightarrow 0$ infrared point by current-type smearing, test functions carrying a p -factor, as for the chiral $U(1)$ current [standard chiral bookkeeping; careful execution in these variables remains to be written out]. On the resulting Weyl algebra take the quasi-free β -KMS state ω_β for the translation flow [existence: standard for this class; the thermal-chiral-net literature is the anchor, cited pending verification]. Then:

- ω_β is KMS for the squeeze flow, *by Proposition 1* — the flow is the same operator in different coordinates;
- ω_β is $SO(3)$ -invariant *automatically*: its covariance is a function of h , which commutes with rotations (Corollary 1.2);
- $W = \hbar \cdot 1$ uniformly: one central scale for all modes, as the centrality theorem licenses;
- nothing is inserted by hand: the algebra came from the tower, the flow from the dilation's $\hbar$ -preserving shadow, the mode continuum from Proposition 1, and the state from the KMS condition itself.

Every constraint of the state problem is met.

5. The type computation

The invariant [standard, Connes]: $S(M) \setminus \{0\}$ is a closed multiplicative subgroup of $\mathbb{R}_+^\times$; the trichotomy $\{1\} / \lambda^\mathbb{Z} / \mathbb{R}_+^\times$ is the type trichotomy semifinite / III $_\lambda$ / III $_1$.

The spectral data [standard; Araki: a quasi-free state is the unique KMS state of its Bogoliubov flow]: for the quasi-free KMS state at inverse temperature β , $\log \text{Spec}(\Delta_\omega)$ is the closed additive group generated by $\beta \cdot \text{supp}(h)$ under finite particle/hole insertions (signs $\pm$).

The computation (verified; grid resolution 10^{-5} , gaps measured on $[-3, 3]$, $\beta = 1$). The generated group was constructed directly for four spectral hypotheses:

spectral data	insertions	max gap	verdict signature
single energy 0.7 (Powers)	4 / 16 / 48	0.700 / 0.700 / 0.700	frozen lattice: III_λ
grading ladder (1, 2, 3) · 0.7	6 / 24 / 48	0.700 / 0.700 / 0.700	frozen lattice: III_λ , $\lambda = e^{-0.7} \approx 0.4966$
two incommensurate $\{1, \sqrt{2}\}$	8 / 24 / 64 / 120	0.243 / 0.071 / 0.029 / 0.017	dense: III_1
interval (0, 1] (our case, forced)	4 / 8 / 16 / 32 energies	0.345 / 0.048 / 10^{-5} / 10^{-5}	saturates $\mathbb{R}$: III_1

The interval row is the only one available to the squeeze-thermal state: Corollary 1.1 forces absolutely continuous spectrum on all of $\mathbb{R}$, so the generated closed group is $\mathbb{R}$ and $\text{Spec}(\Delta_\omega) = [0, \infty)$.

The one cited link. $\text{Spec}(\Delta_\omega) = S(M)$ requires the centralizer condition [Connes' criterion], standard for free thermal states [Araki-Woods 1963; Araki-Woods, *A classification of factors*; Bratteli–Robinson II]. This is the single link of the chain not re-derived here, and it is flagged, not smuggled.

Verdict. Conditional on that link: the factor of the squeeze-thermal state is of type III_1 — and by Haagerup's uniqueness theorem, it is *the* hyperfinite III_1 factor. The destination is canonical; the octonionic origin lives entirely in the labeling: the flow that is thermal, the rotational symmetry the state respects, the scale line that became the mode space, and the $O(\mu)$ R-flux corrections as perturbative memory.

6. The fork, resolved into two theorems

The question left deliberately open throughout this series' preparation phase — type III_1 or type III_λ — does not dissolve; it splits into two precise statements, both now computed. *Within* the squeeze-KMS formulation, III_1 is forced: the spectrum is an interval by Proposition 1, and intervals saturate. **Outside** it, the alternative is exactly characterized: a state coupled instead to the discrete grading ladder — the octonionic weights (1, 2, 3) are commensurate — yields type III_λ with $\lambda = e^{-\beta E_0}$, a preferred scale in the modular spectrum, set by the contraction. The program's claim is therefore conditional in exactly one place, and that conditionality is its falsifiable content: ***if* thermal time runs along the inherited dilation flow, the algebra of the world is the unique hyperfinite type III_1 factor; if nature instead couples to the grading ladder, the modular spectrum carries a preferred octonionic scale.**** Either outcome remembers the origin.

7. Scope

Three items remain outside what is established here, stated precisely. (i) The centralizer

condition of §5 is cited, not re-derived, for our specific chiral construction; verifying it against the standard sources is the remaining step of the type argument. (ii) Uniqueness of ω_β among $SO(3)$ -invariant squeeze-KMS states is open; the thermal-chiral classification literature likely decides it. (iii) The infrared execution (current-type smearing in these variables) is standard in kind but has not been written out in detail. One naming caution for the record: Shlyakhtenko's *free* Araki-Woods factors — a distinct, free-probability construction — exhibit the same lattice-versus-line type trichotomy through their orthogonal representation's spectral data; this is cited as a parallel instance of the mechanism, not as evidence for the present construction. Finally, bibliographic details below are given only to the precision verified within this series' sessions; page numbers are included only where independently confirmed.

References

- Companion papers: *The Second Contraction: From Algebraic R-Flux to the Heisenberg Algebra, the Dilation Symmetry It Creates, and the $SO(3)$ Remnant of G_2 ; The Central Element of Algebraic R-Flux.*
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